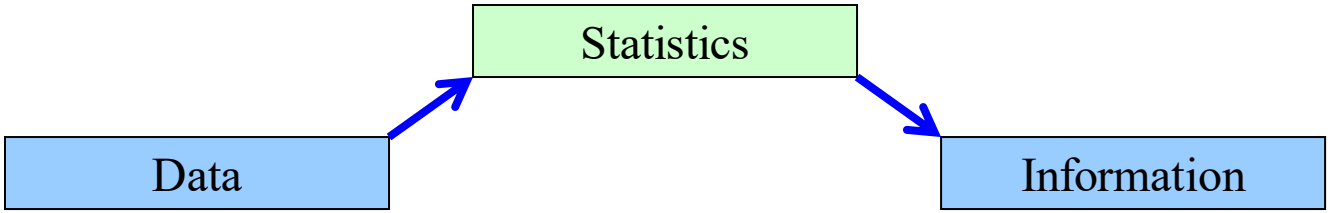
Econometrics

Basic Concepts in Statistics

What is Statistics?

- Statistics is the science of collecting, organizing, analyzing, interpreting, and presenting data.

"Statistics is a way to get information from data"



Data: Facts, especially numerical facts, collected together for reference or information.

Information: Knowledge communicated concerning some particular fact.

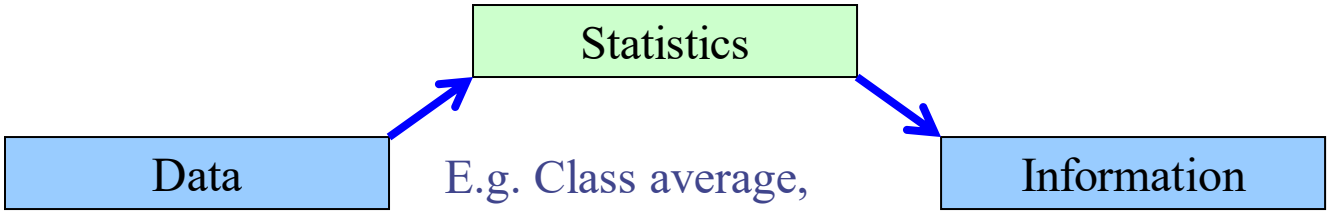
Statistics is a ***tool*** for creating ***new understanding*** from a set of numbers.

Definitions: Oxford English Dictionary

What is Statistics?

- Statistics is the science of collecting, organizing, analyzing, interpreting, and presenting data.

"Statistics is a way to get information from data"



List of last terms' scores:
95, 82, 70, 54, 78, 69, 89,…

E.g. Class average, Proportion of class receiving A's, Most frequent mark, Marks distribution, etc.

New information about the class.

Statistics is a ***tool*** for creating ***new understanding*** from a set of numbers, it gives us good descriptions about the object.

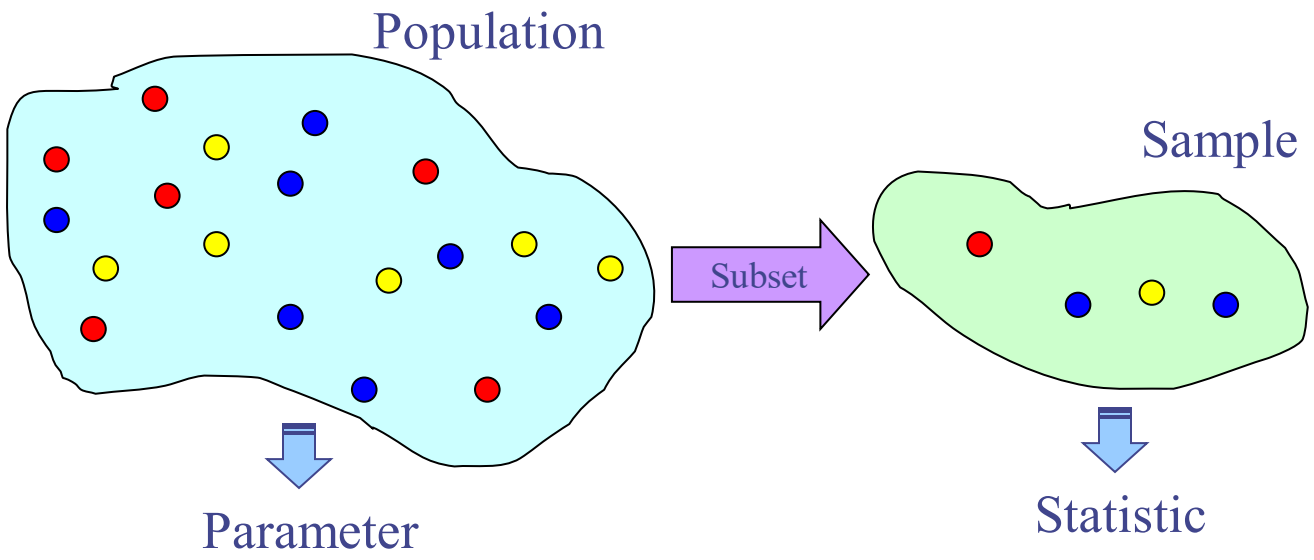
Definitions: Oxford English Dictionary

Population vs. Sample

- Population
  - a population is the group of all items of interest to a statistics practitioner.
  - frequently very large, sometimes infinite.

- Sample
  - a sample is a set of data drawn from the population.
  - potentially very large, **but less than the population**.

Key Statistical Concepts



Population has Parameter

Sample has Statistic

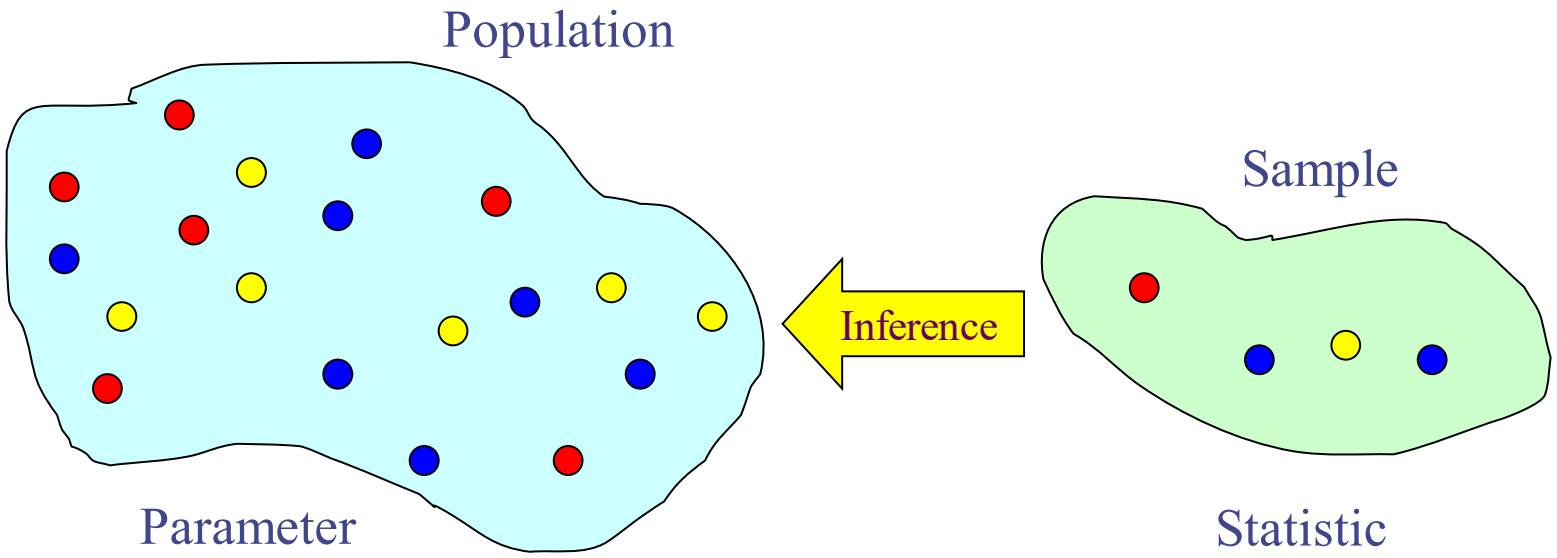
- A statistic is a single measure (number) used to summarize a sample data set.

Descriptive vs. Inferential Statistics

- **Descriptive statistics:** involves arranging, summarizing, and presenting a set of data in such a way that useful information is produced.
  - Its methods make use of graphical techniques and numerical descriptive measures (such as averages) to summarize and present the data
- **Inferential statistics:** generalizing ***from a sample to a population***, estimating unknown population parameters, drawing conclusions, making decisions.

Statistical Inference

- **Statistical inference** is the *process* of making an estimate, prediction, or decision about a population based on a sample.



We can ***infer*** about a Population's Parameters based on a Sample's Statistics.

Notations

- When referring to the number of observations in a ***population***, we use uppercase letter N.
- When referring to the number of observations in a ***sample***, we use lower case letter n.
- The mean for a ***population*** is denoted with μ : Parameter or Statistic?
- The mean for a ***sample*** is denoted with $\bar{x}$: Parameter or Statistic?

Outline of Dispersion

- *Dispersion* is the "spread" of data points about the center of the distribution.
- *Measures of dispersion:*
  - Variance
  - Standard deviation

Variance

- The population variance is defined as the sum of squared deviations around the mean divided by the population size.

$$\sigma^2 = \frac{\sum_{i=1}^{n}(x_i - \mu)^2}{N}$$

Variance

- For the sample variance, however, we divide by $n - 1$ instead of n, otherwise it would tend to underestimate the unknown population variance.

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

Standard Deviation

- The square root of the variance, which explains how individual values in a data set vary from the mean.
- Population SD:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \mu)^2}{N}}$$

- Sample SD:

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

Calculating a Standard Deviation

- Consider the following five quiz scores.

i	x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	x_i^2
1	40	$40 - 72 = -32$	$(-32)^2 = 1024$	$40^2 = 1600$
2	55	$55 - 72 = -17$	$(-17)^2 = 289$	$55^2 = 3025$
3	75	$75 - 72 = +3$	$(3)^2 = 9$	$75^2 = 5625$
4	95	$95 - 72 = +23$	$(23)^2 = 529$	$95^2 = 9025$
5	95	$95 - 72 = +23$	$(23)^2 = 529$	$95^2 = 9025$
Sum	360	0	2380	28300
Mean	72			

- Now, calculate the sample standard deviation:

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{2380}{5-1}} = \sqrt{595} = 24.39$$

Calculating a Standard Deviation

- The standard deviation is nonnegative because deviations around the mean are squared.
- When every observation is exactly equal to the mean, the standard deviation is zero.
- Standard deviations can be large or small, depending on the units of measure.
  - Thus, compare standard deviations only for data sets measured in the same unit.

Measures of dispersion

	Population	Sample
Size	N	n
Mean	μ	$\bar{x}$
Variance	$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$	$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$
Standard Deviation	σ	s

Conclusion

- Population vs. sample
- Measure of dispersion: variance and standard deviation

- Next time: review probability